

Dear Hiring Team,

I am a Ph.D. chemist at Texas A&M University with over four years of experience spanning organic synthesis, high-throughput experimentation, computational modeling, and mechanistic analysis. My research combines synthetic design, quantum chemical modeling, and data-driven experimentation to address challenging problems in chemical reactivity and reaction development. This interdisciplinary training has prepared me to contribute effectively to both process chemistry and small-molecule research programs.

During graduate school, I have developed and optimized novel transition-metal-catalyzed transformations with a strong emphasis on mechanistic understanding, reaction design, and multistep synthesis. I have extensive hands-on experience working with air-sensitive iron complexes and troubleshooting challenging synthetic pathways, as well as characterizing complex intermediates and final products. This background has strengthened my ability to solve complex experimental problems, work across disciplines, and contribute productively in collaborative research environments.

My doctoral research focused on the development of asymmetric iron-catalyzed multicomponent cross-coupling reactions that achieve greater than 98% enantio- and diastereoselectivity within one hour, representing a significant improvement over prior approaches. To further advance this work, I conducted research in Professor Michael Neidig's laboratory at the University of Oxford in Fall 2023, where I gained hands-on experience with Mössbauer spectroscopy to identify key reactive iron intermediates and investigate stereodetermining steps. These studies provided new mechanistic insight into iron catalysis and strengthened my ability to use mechanistic understanding to guide hypothesis-driven reaction development.

Throughout my Ph.D., I have built a broad collaborative network, working with more than 10 academic research groups on projects spanning photochemistry, transition-metal catalysis, and inorganic synthesis. These collaborations involved Pd, Ni, Fe, Cu, and Rh systems and required effective cross-disciplinary communication and the integration of diverse technical perspectives. As a result, I have developed the ability to communicate complex scientific ideas clearly, adapt quickly to new research settings, and contribute effectively within collaborative teams.

At UCLA, as a Visiting Graduate Researcher, I have been developing iron-catalyzed reductive cross-coupling strategies for asymmetric alkene dicarbofunctionalization. By leveraging UCLA's high-throughput experimentation platform, I have carried out rapid chiral ligand screening and reaction optimization to accelerate discovery through data-rich experimentation. This experience further strengthened my ability to integrate synthesis, computation, and parallel experimentation to solve challenging chemical problems efficiently.

I have authored **21 peer-reviewed publications**, including two first-author and eight second-author papers, and have presented my research at numerous international conferences, including ACS Fall 2024, the GRC on Physical Organic Chemistry, the CIC Annual Meeting, and UCLA's Houk Conference. My work has been recognized through several honors, including the **Sharon Dabney Memorial Scholarship** for research excellence and departmental service, as well as the **Third Best Poster Award** at the 2023 SACNAS Diversity in Science Symposium.

Beyond research, I have demonstrated leadership through mentoring, outreach, and service. I have mentored eight students and co-led the iCarbon initiative, which introduced computational chemistry to underrepresented community college students. I also serve as Secretary of Phi Lambda Upsilon (PLU),

contribute to STEM outreach and recruitment efforts, and represent the Physical Chemistry Division on the Chemistry Student Safety Committee, where I help organize departmental Safety Roundtable Talks.

Key Highlights:

- Developed iron-catalyzed asymmetric multicomponent and reductive cross-coupling methodologies integrating HTE and DFT for sustainable catalysis.
- Performed quantum chemical simulations to guide experimental design and elucidate mechanistic pathways of multiple organometallic transformations.
- Collaborated internationally across 10+ institutions, demonstrating adaptability and strong communication in multidisciplinary settings.
- Authored **21** peer-reviewed publications, presented at major conferences, and received recognition for research and leadership excellence.
- Mentored students, led outreach initiatives, and managed laboratory instrumentation including gloveboxes, HPLC systems, and spectroscopy tools.

I am particularly drawn to research environments that value rigorous problem-solving, collaboration, and the integration of mechanistic insight with experimental innovation. I see this next stage of training as an opportunity to deepen my expertise, broaden my research perspective, and continue developing the independent scientific foundation needed for a long-term academic career, including future faculty opportunities at leading research and teaching institutions in India.

Thank you for your time and consideration.

Achyt Ranjan Gogoi